## Homework

**DM1 - A sum and a threshold - 45 min, after S11.**

Consider the sequence $(u_n)$ defined by $u_0 = 2$ and, for every $n \in \mathbb{N}$, $u_{n+1} = \frac{1}{2}u_n + 3$.

1. Prove by induction that $2 \leq u_n < 6$ for all $n \in \mathbb{N}$. Deduce the direction of variation of $(u_n)$.

2. Let $v_n = u_n - 6$ for all $n \in \mathbb{N}$. Show that $(v_n)$ is geometric, then express $u_n$ in function of $n$.

3. For every $n \in \mathbb{N}$, let $S_n = \sum_{k=0}^{n} u_k$. Express $S_n$ as a function of $n$.

4. Using the explicit formula and powers of 2, determine the smallest integer $n$ such that $u_n > 5.99$. Check that the previous integer does not work.

5. Express $6 - \frac{S_n}{n+1}$ and justify that this difference is strictly positive. Interpret the result in considering the mean of the first $n + 1$ terms; no limit calculation is required.